

## **PROJECT PROPOSAL**

### **Smart Attendance and Classroom Access Platform with Face Verification and Geofenced Check-In**

PID 12

Mentor: Mr. Pasan Sarathchandra

Group 27

|         |                           |
|---------|---------------------------|
| 230702K | WICKRAMAARACHCHI W.A.A.U. |
| 230714A | WIJAYARATHNE T.W.O.A.     |
| 230734J | WITHANA W.D.P.S.          |
| 230736R | WITHANAGE M.H.            |

Name : Withanage M.H

Index : 230736R

### 1. Title of the project :

Smart Attendance and Classroom Access Platform with Face Verification and Geofenced Check-In

### 2. Overview of the Project:

This project proposes a web and mobile-based secure attendance management platform for universities. Lecturers create and manage attendance sessions through a web application, while students mark their attendance through a mobile application. Each attendance check-in verifies the student's identity and location to prevent proxy attendance.

*Inputs:* Student registration details and facial images sourced from existing university student records, facial images captured during attendance check-in, device location data for geofence validation and dynamic QR code scanning for classroom presence confirmation, lecturer defined class sessions and timetables and administrator configured geofence boundaries for classrooms or lecture locations.

*Outputs:* Verified attendance records, live attendance status during active sessions, student attendance history, attendance reports and analytics, and notifications for attendance related alerts and upcoming classes.

### 3. Objectives of the Project :

- Design and implement a mobile and web-based attendance management system.
- Allow lecturers to create, activate, and close attendance sessions for lectures.
- Allow students to mark attendance through a mobile application.
- Verify student identity using backend face verification based on face embeddings.
- Reduce proxy attendance using face verification and location based validation.
- Generate attendance reports for lecturers, administrators, and students.
- Provide a centralized system to manage students, lecturers, courses, and attendance records.

### 4. The Need for the Project :

Attendance is an important part of academic management in universities and training institutions. However, many institutions still use paper based or basic digital attendance methods. These methods can be slow, error-prone, and difficult to manage when there are large student groups.

A smart attendance system is useful because it saves lecture time, reduces manual work, improves accuracy, and provides better access to attendance records. It also helps lecturers identify low attendance early and allows lecturers and administrators to generate reports without manually checking attendance sheets. Since attendance fraud such as proxy marking can occur in conventional systems, identity verification and location validation are essential for improving the reliability of attendance records.

However, single factor verification methods also have limitations. GPS only attendance can be unreliable indoors, QR code only attendance can be shared with absent students, and face verification alone cannot confirm that a student is physically present in the classroom. Therefore, combining face verification with geofenced location validation and dynamic QR code verification makes the system more reliable.

### 5. Scope of the Project :

*Students (Mobile Application)*

- Log in and maintain a personal profile.
- View personal profile, timetable, and attendance history.
- Mark attendance by performing face verification and location validation.
- Receive notifications for upcoming classes and attendance related alerts.

*Lecturers (Web Application):*

- Log into the system and maintain a personal profile.
- Create and manage courses and attendance sessions, timetables.
- View live attendance status during active sessions.
- Manually mark attendance for verification failed students.

- Generate attendance reports and monitor students attendance analytics.

#### *Administrators (Web Application):*

- Create and manage student and lecturer accounts.
- Configure classrooms, geofence locations, and system settings.
- View university wide attendance reports and analytics.
- Manage courses and academic information.

#### *Main Functions of the System*

- User authentication with role based access control.
- Face verification during attendance check-in.
- Geofenced and dynamically refreshing QR code based location validation to confirm classroom presence.
- Attendance session management.
- Attendance history, reporting, and analytics.
- Secure storage of attendance records and verification logs.

#### 6. Deliverables :

A web-based attendance management system and a mobile application that enable secure attendance management for lecturers, administrations, and students through face verification and geofenced check-ins. The system will provide intuitive graphical user interfaces, a secure backend with REST APIs and authentication, attendance reporting and analytics dashboards at university, class and individual student level, and an AI based face verification module built on a pretrained face embedding model, integrated into the check-in workflow.

The project will also deliver technical documentation including the system architecture, database design, API design, and AI based face verification workflow along with a user guide for system operation, to support deployment, maintenance, and future enhancements.

#### 7. Overview of Existing Systems and Technology :

Jibble and Truein are employee based attendance systems that use face recognition with GPS tracking and geofencing. However these systems are mainly useful for field staff and multi-site environments [1],[2]. Qwickly Attendance and SEAtS ONE are more related to educational institutions. They support methods such as QR code scanning, location confirmation, Bluetooth beacons, Wi-Fi and card based check-in for student attendance [3],[4].

AttenFace is a research based classroom attendance system that uses classroom camera snapshots and face recognition [5].

A closely related study by Babu et al. [6] combines geofencing with facial recognition. GPS location is validated before a facial embedding, generated via the face\_recognition library, is matched against a stored reference on a Django/MySQL backend. Liveness detection is not yet implemented and noted as future work. This project adopts the same dual verification principle, with location validation via GPS based geofencing and a time limited, dynamically refreshing QR code.

#### Planned Technology Stack:

Web Application - Next.js and TypeScript [7]

Student Mobile Application - React Native with Expo [8]

Backend - Python FastAPI [9]

Database - PostgreSQL [10]

Geofenced Location Validation - GPS coordinate and radius based validation

Face Embedding Extraction - DeepFace [11] using a pretrained ArcFace model [12]

Similarity Comparison - Cosine similarity with a configurable verification threshold [13]

Face Landmark Detection and Liveness Check - Google ML Kit face detection API [14]

Image Preprocessing - OpenCV [15]

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